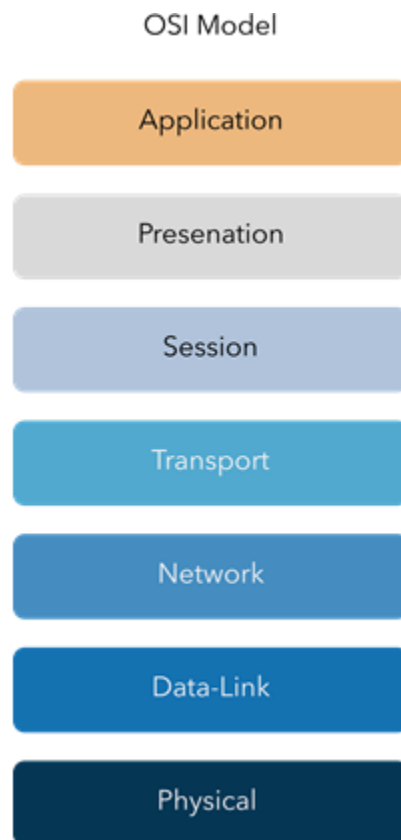


OSI Layer:-

Tasks :-

1. Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).

Ans. :-



OSI Layer	Purpose	Example Device/Protocol
7. Application	Provides network services to end-user applications	HTTP, FTP, SMTP, DNS
6. Presentation	Translates, encrypts, and compresses data	SSL/TLS, JPEG, ASCII
5. Session	Establishes and manages communication sessions	NetBIOS, RPC
4. Transport	Ensures reliable data delivery and error recovery	TCP, UDP
3. Network	Routes packets between networks	Router, IP
2. Data Link	Transfers data between devices on the same network	Switch, Ethernet (MAC)
1. Physical	Transmits raw bits over physical media	Hub, Ethernet cable, Fiber optic cable

2. Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames.

Hint: Write your answers as 'Scenario: Layer Name (Layer Number)'.

Ans. :-

Scenario	OSI Layer	Reason
a) Encrypting data before sending	Presentation Layer (Layer 6)	Handles data encryption, decryption, compression, and formatting.
b) Routing packets across networks	Network Layer (Layer 3)	Responsible for routing packets between different networks using IP addresses.
c) Converting data to electrical signals	Physical Layer (Layer 1)	Converts digital data into electrical, optical, or radio signals for transmission.
d) Opening a web page in your browser	Application Layer (Layer 7)	Uses protocols like HTTP and HTTPS to provide web services to users.
e) Detecting and correcting errors in frames	Data Link Layer (Layer 2)	Detects transmission errors in frames using mechanisms such as CRC and manages reliable communication on the local network.

3. Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

Ans. :-

Example: WhatsApp :-

When I send a message on WhatsApp, multiple OSI layers work together to deliver the message successfully.

1. Application Layer (Layer 7) :-

- This is where WhatsApp operates. When I type a message and tap Send, the app creates the data and communicates using internet protocols such as HTTPS.

2. Presentation Layer (Layer 6) :-

- Before the message is transmitted, WhatsApp encrypts it using end-to-end encryption so that only the sender and receiver can read the message.

3. Transport Layer (Layer 4):-

- The TCP protocol ensures that the message is delivered reliably, in the correct order, and without missing data.

4. Network Layer (Layer 3):-

- The IP protocol and routers determine the best path for the message to travel from my phone to WhatsApp's servers and then to the recipient's device.

5. Data Link Layer (Layer 2):-

- My phone sends the data over Wi-Fi or mobile data. The network interface uses MAC addresses to communicate with the local router or cellular tower.

6. Physical Layer (Layer 1):-

- Finally, the message is transmitted as electrical, radio, or optical signals through Wi-Fi, mobile networks, and internet cables.

When I send a WhatsApp message, the Application Layer creates the message, the Presentation Layer encrypts it, the Transport Layer ensures reliable delivery, the Network Layer routes it, the Data Link Layer transfers it over the local network, and the Physical Layer carries it as signals to the destination.

4. Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

Ans. :-

If you can successfully ping the website's IP address but cannot open the website in a browser, the lower network layers are working correctly. The problem is most likely at the upper OSI layers.

Possible OSI Layer	Reason
Application Layer (Layer 7)	The web service (HTTP/HTTPS) may be down, the web server may not be responding, or the browser may have an issue.
Presentation Layer (Layer 6)	If the website uses HTTPS, there could be an SSL/TLS encryption or certificate problem preventing the browser from establishing a secure connection.
Session Layer (Layer 5)	There may be a problem establishing or maintaining the communication session between the browser and the web server.

Reasoning :-

- Ping uses ICMP, which operates at the Network Layer (Layer 3). Since the ping is successful, the Physical (Layer 1), Data Link (Layer 2), and Network (Layer 3) layers are functioning correctly.
- The issue occurs only when trying to access the website, which uses HTTP or HTTPS at the Application Layer (Layer 7).
- If the website uses HTTPS, an SSL/TLS issue at the Presentation Layer (Layer 6) could also prevent access.
- In some cases, the Session Layer (Layer 5) may fail to establish or maintain the connection.

The problem is most likely at the Application Layer (Layer 7), and it could also involve the Presentation Layer (Layer 6) or Session Layer (Layer 5). Since the website's IP address responds to ping, the lower OSI layers are working properly, and the fault is likely with the web service, browser, or secure communication rather than the network connection itself.

